

Far-Left Stability of a Chemotaxis Front

The sharp threshold $\chi = 4$, a localized entropy proof, and the necessary basin hypothesis

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Abstract

For the normalized one-dimensional chemotaxis–logistic equation

$$u_t = u_{zz} + (c - \chi v_z)u_z + u(1 - \chi v + (\chi - 1)u), \quad -v_{zz} + v = u,$$

we determine the exact stability threshold of the plateau $(u, v) = (1, 1)$ behind a traveling front. The linear dispersion relation is strictly negative for every Fourier mode exactly when $0 \leq \chi < 4$; at $\chi = 4$ the mode $k = 1$ is neutral, and for $\chi > 4$ an unstable band appears. A nonlinear relative-entropy calculation has the same sharp coefficient $1 - \chi^2/16$. We localize that calculation on the whole line with a slowly varying weight and explicit boundary errors, and obtain uniform far-left convergence throughout $0 \leq \chi < 4$, conditional on a persistent positive far-left band and the stated derivative compactness.

The basin condition cannot be removed for the natural broad class of front-like solutions. A genuine traveling wave of speed s , observed in a faster frame $c > s$, is positive at the left end on every time slice but converges to zero at every fixed point of that frame. This convective-escape example is formalized for an actual PDE orbit already at $\chi = 0$. All theorem and counterexample statements cited below are checked in Lean 4.30.0 / Mathlib v4.30.0, without `sorry`; their axiom audits use only `propext`, `Classical.choice`, and `Quot.sound`.

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1 Problem and normalization

In the traveling-wave problem the original population–signal system is placed in the frame $z = x - ct$. In the normalized case

$$m = \alpha = \gamma = 1$$

the equation can be written in either of two equivalent ways:

$$u_t = u_{zz} + (c - \chi v_z)u_z + u(1 - \chi v + (\chi - 1)u), \quad -v_{zz} + v = u, \quad (1)$$

$$u_t = u_{zz} + cu_z - \chi(uv_z)_z + u(1 - u), \quad -r_{zz} + r = u - 1, \quad r = v - 1. \quad (2)$$

Indeed $v_{zz} = v - u$, so expanding the divergence in (2) gives (1).

The traveling profile connects

$$(u, v) = (1, 1) \quad \text{at } z = -\infty \quad \text{to} \quad (u, v) = (0, 0) \quad \text{at } z = +\infty.$$

The question addressed here is not construction of the wave, but stability of the left plateau in the moving frame. The target conclusion is the quantified statement

$$\forall \varepsilon > 0 \exists R, T \forall t \geq T \forall z \leq -R, \quad |u(t, z + ct) - 1| < \varepsilon. \quad (3)$$

In Lean this is `UniformCoMovingLeftEquilibriumConvergencecorbit`.

2 The exact linear threshold

2.1 Dispersion relation

Write

$$u = 1 + p, \quad v = 1 + r.$$

The elliptic equation linearizes exactly to

$$-r_{zz} + r = p.$$

The product $r_z p_z$ is quadratic, while the linearization of the reaction term in (1) is $(\chi - 1)p - \chi r$. Hence

$$p_t = p_{zz} + cp_z + (\chi - 1)p - \chi r.$$

For a Fourier mode $p = e^{\lambda t + i k z}$, $s = k^2 \geq 0$, the resolver gives

$$\hat{r} = \frac{\hat{p}}{1 + s},$$

and therefore

$$\lambda(s) = -s + i c k + (\chi - 1) - \frac{\chi}{1 + s}. \quad (4)$$

The transport $i c k$ affects only the imaginary part. The real growth rate has the useful form

$$d_\chi(s) := \operatorname{Re} \lambda(s) = -1 - s + \frac{\chi s}{1 + s}. \quad (5)$$

2.2 Maximization

There are two regimes. If $0 \leq \chi \leq 1$, then

$$d'_\chi(s) = -1 + \frac{\chi}{(1+s)^2} \leq 0,$$

so the maximum on $s \geq 0$ is $d_\chi(0) = -1$. If $\chi > 1$, the unique critical point is

$$s_* = \sqrt{\chi} - 1 > 0,$$

and it is the global maximizer. Substitution gives

$$d_\chi(s_*) = (\sqrt{\chi} - 1)^2 - 1 = \chi - 2\sqrt{\chi}. \quad (6)$$

Consequently

$$d_\chi(s) < 0 \quad \text{for all } s \geq 0 \quad \Longleftrightarrow \quad 0 \leq \chi < 4.$$

At $\chi = 4$, $s_* = 1$, hence $k_* = \pm 1$, and the maximal real part is zero. For $\chi > 4$, the maximizing mode has positive growth.

The formalized general scalar dispersion is

$$d_{\alpha, \chi\gamma}(s) = -\alpha - s + \frac{\chi\gamma s}{1+s}.$$

For $\chi\gamma > 1$, its maximum is

$$(\sqrt{\chi\gamma} - 1)^2 - \alpha,$$

so the exact threshold is

$$\chi\gamma < (1 + \sqrt{\alpha})^2. \quad (7)$$

For $\chi\gamma \leq 1$, the maximum is the zero mode $-\alpha$, which is already negative.

Theorem 1 (sharp dispersion threshold). *Let $\alpha > 0$ and $\chi\gamma \geq 0$. If $\chi\gamma < (1 + \sqrt{\alpha})^2$, then $d_{\alpha, \chi\gamma}(s) < 0$ for every $s \geq 0$. If $\chi\gamma > (1 + \sqrt{\alpha})^2$, then some $s \geq 0$ has positive growth. When $\chi\gamma > 1$, the maximum is attained at $s = \sqrt{\chi\gamma} - 1$.*

The corresponding Lean theorems are `dispersion_le_of_lt_turing`, `dispersion_pos_of_gt_turing`, and `dispersion_attains_at_sqrt` in `Paper1/WholeLineChiPosDispersionSharp.lean`.

3 The sharp nonlinear entropy identity

3.1 Relative entropy and the exact cancellation

For $u > 0$, define

$$h(u) = u - 1 - \log u, \quad h'(u) = 1 - \frac{1}{u}.$$

Let

$$A = \frac{u_z}{u}, \quad W = u - 1.$$

On a periodic interval, or before boundary terms are restored, multiply (2) by $h'(u)$ and integrate. The diffusion term is

$$\int h'(u) u_{zz} = - \int A^2.$$

The constant drift is a total derivative and integrates to zero. For the chemotaxis term,

$$\begin{aligned} -\chi \int h'(u)(uv_z)_z &= -\chi \int \left(1 - \frac{1}{u}\right) (u_z r_z + u r_{zz}) \\ &= \chi \int A r_z. \end{aligned}$$

The second equality follows by integrating the $W r_{zz}$ term by parts; this is the cancellation that keeps the logarithmic entropy sharp. Finally,

$$\int h'(u)u(1-u) = -\int W^2.$$

Thus

$$\frac{d}{dt} \int h(u) = -\int A^2 + \chi \int A r_z - \int W^2. \quad (8)$$

Young's inequality gives

$$\chi A r_z \leq A^2 + \frac{\chi^2}{4} r_z^2.$$

The coefficient 1/4 in the one-dimensional elliptic resolver estimate is also sharp:

$$\int r_z^2 \leq \frac{1}{4} \int W^2, \quad -r_{zz} + r = W. \quad (9)$$

In Fourier variables, the multiplier is

$$\frac{k^2}{(1+k^2)^2} \leq \frac{1}{4},$$

with equality at $k^2 = 1$. Combining (8)–(9) yields

$$\frac{d}{dt} \int h(u) \leq -\left(1 - \frac{\chi^2}{16}\right) \int (u-1)^2. \quad (10)$$

The coefficient is positive exactly for $|\chi| < 4$. In the attractive range considered here, this is precisely $0 \leq \chi < 4$, matching the linear threshold and the critical mode.

Equation (10) is formalized as `sharp_entropy_production_le` in `Paper1/WholeLineChiPosSharpEntropyDissipation.lean`. The proof is genuinely nonlinear: it assumes positivity of u , not smallness of $u-1$.

3.2 Why Fourier analysis is not enough on the moving far left

The target (3) is a whole-line, moving-half-line statement. A periodic calculation has no boundary flux, while a sharp Fourier multiplier does not directly localize to a moving window. The formalization therefore proves the resolver estimate in real space with a positive weight w satisfying

$$|w'(z)| \leq \frac{1}{\ell} w(z), \quad \ell > 0. \quad (11)$$

On a finite interval $[a, b]$, integration by parts gives

$$\begin{aligned} \int_a^b w r_z^2 &\leq \frac{1}{4} \int_a^b w W^2 + \frac{1}{2\ell} \left(\int_a^b w r^2 + \int_a^b w r_z^2 \right) \\ &\quad + |w(b)r(b)r_z(b) - w(a)r(a)r_z(a)|. \end{aligned} \quad (12)$$

The leading source coefficient remains exactly 1/4; every loss is displayed as a slow-weight or endpoint term.

3.3 The localized production inequality

Applying the same weighted integration by parts to the population equation gives

$$\begin{aligned} \int_a^b wh'(u)u_t \leq & - \left(1 - \frac{\chi^2}{16}\right) \int_a^b wW^2 \\ & + \frac{|c| + \chi + 2}{\ell} \left[\int_a^b wh(u) + \int_a^b w(A^2 + W^2) \right] \\ & + \left(\frac{\chi^2}{4} + \frac{\chi}{2\ell} \right) \left[\frac{1}{2\ell} \left(\int_a^b wr^2 + \int_a^b wr_z^2 \right) + B_{\text{res}} \right] + B_{\text{pop}}, \end{aligned} \quad (13)$$

where the boundary terms are explicit:

$$\begin{aligned} B_{\text{res}} &= |w(b)r(b)r_z(b) - w(a)r(a)r_z(a)|, \\ B_{\text{pop}} &= [wh'(u)u_z]_a^b + c[wh(u)]_a^b - \chi[wWr_z]_a^b. \end{aligned}$$

No unspecified $O(1/\ell)$ term hides dependence on the solution. The complete inequality is `weighted_sharp_entropy_production_le`, and (12) is `weighted_resolver_le`, both in `Paper1/WholeLineChiPosWeightedEntropyDissipation.lean`. The instantiation on the canonical Cauchy orbit is `wholeLineCauchyGlobalU_weighted_sharp_dissipation`; the literal time derivative of the entropy window is proved in `Paper1/WholeLineChiPosWeightedEntropyOrbitDeriv.lean`.

4 From entropy to far-left convergence

4.1 The two nontrivial hypotheses

The theorem uses a persistent far-left band

$$\exists T, Z \forall t \geq T \forall z \leq Z, \quad \ell \leq u(t, z + ct) \leq M, \quad \ell > 0, \quad (14)$$

formalized as `EventualCoMovingLeftBandcellMorbit`. The upper bound is available from the global ceiling. The important part is the time-uniform positive floor.

It also assumes `ChiOneFarLeftDerivativeCompactness`: every sequence of time shifts $t_n \rightarrow +\infty$ and far-left space shifts $z_n \rightarrow -\infty$ carries the first- and second-order population/resolver fields with the uniform estimates needed to extract a classical translated limit. This is the precise compactness input; the theorem does not hide it inside a statement called “regularity.”

4.2 Liouville/compactness mechanism

The proof follows four steps.

1. Suppose (3) fails. Then some $\varepsilon_0 > 0$ and sequences $t_n \rightarrow \infty$, $z_n \rightarrow -\infty$ satisfy $|u(t_n, z_n + ct_n) - 1| \geq \varepsilon_0$.
2. Translate the orbit by (t_n, z_n) . The band (14) gives uniform positive lower and upper bounds on every fixed compact box. Derivative compactness extracts an entire classical limit.
3. Apply the localized entropy inequality on successively larger windows. The slow-weight and endpoint errors vanish in the exhaustion. Since $1 - \chi^2/16 > 0$, the entire limit has zero displacement dissipation: $\int (u - 1)^2 = 0$ on every compact interval.

4. Hence the limit is $u \equiv 1$, contradicting the nontrivial value at the translated origin.

Theorem 2 (basin-conditional sharp-range convergence). *Let $0 \leq \chi < 4$, $\ell > 0$, and let an orbit satisfy `EventualCoMovingLeftBandcellMorbit` and `ChiOneFarLeftDerivativeCompactnesschicorbit`. Then `UniformCoMovingLeftEquilibriumConvergencecorbit`; equivalently, (3) holds.*

This is `uniformCoMovingLeftEquilibriumConvergence_chiPos_upto_four_basinConditional` in `Paper1/WholeLineChiPosEntropyFarLeftBasinConditional.lean`. The same file contains the canonical-orbit version. A separate theorem derives the band from a persistent lower-barrier plateau and the global ceiling, leaving only the derivative compactness input.

5 Why the basin floor cannot be omitted

5.1 The quantifier distinction

Positivity at the left endpoint on each time slice says

$$\forall t \exists \ell_t > 0 \exists Z_t \forall z \leq Z_t, \quad u(t, z + ct) \geq \ell_t. \quad (15)$$

The persistent band needed above says

$$\exists \ell > 0 \exists T, Z \forall t \geq T \forall z \leq Z, \quad u(t, z + ct) \geq \ell. \quad (16)$$

The order of the first three quantifiers is reversed. Classical positivity on fixed slices does not imply (16).

5.2 Convective escape

Let (U, V) be a genuine traveling wave of laboratory speed s :

$$u_{\text{lab}}(t, x) = U(x - st), \quad v_{\text{lab}}(t, x) = V(x - st),$$

with $U(-\infty) = 1$ and $U(+\infty) = 0$. Observe it in a frame of speed $c > s$. The population profile is

$$q(t, z) = u_{\text{lab}}(t, z + ct) = U(z + (c - s)t). \quad (17)$$

For every fixed t ,

$$\lim_{z \rightarrow -\infty} q(t, z) = 1,$$

so (15) holds. On the other hand, for every fixed z ,

$$z + (c - s)t \longrightarrow +\infty, \quad q(t, z) \longrightarrow 0.$$

Therefore, for every proposed $\ell > 0, T, Z$, one can choose $t \geq T$ so that $q(t, Z) < \ell$. No persistent floor exists.

This is a speed mismatch, not a failure of pointwise positivity and not an artifact of a fabricated trajectory. The chain rule verifies that (q, W) , with $W(t, z) = V(z + (c - s)t)$, solves the exact moving-frame PDE. The repository constructs a concrete example at

$$\chi = 0, \quad s = 3, \quad c = 4.$$

Proposition 1 (genuine PDE obstruction). *There exists a global classical traveling-wave orbit at $\chi = 0$ such that every fixed co-moving time slice is strictly positive at the left endpoint, but no positive *EventualCoMovingLeftBand* exists in the faster frame.*

The pure quantifier theorem is `convectiveEscapeProfile_no_uniform_left_floor`; the wave-level statement is `IsTravelingWave.convectiveEscape_quantifier_obstruction`; and the constructed $\chi = 0$ PDE counterexample is `exists_genuine_convectiveEscape_counterexample_chi_zero` in `Paper1/WholeLineConvectiveEscapePDE.lean`.

5.3 What the counterexample does and does not refute

It refutes an unconditional theorem quantified over broad front-like data and an arbitrary observation speed. It does not refute the paper’s wave-stability theorem with a weighted initial-closeness condition to a fixed wave. Such a condition pins the orbit and the frame to the same wave speed; the convective-escape orbit is not close to the incorrectly framed wave in the required weighted norm.

Thus the optimal sharp-range statement has one of two equivalent kinds of extra information:

- a persistent plateau-band hypothesis such as (14); or
- a wave-closeness/phase condition strong enough to rule out speed mismatch and imply that band.

6 Audited alternative routes

Several plausible approaches were developed far enough to identify their exact obstruction. They are recorded here to distinguish a failed estimate from an untried idea.

Localized L^1 mass. For a cutoff of width L , $|\phi'_L| = O(L^{-1})$, but the transition annulus has length $O(L)$. With only a uniform bound on the orbit, the transport/chemotaxis leakage is $O(1)$, not $o(1)$. Increasing the window therefore does not prevent all mass from convecting through its boundary. The escape profile (17) realizes exactly this mechanism.

Bare L^2 deficit energy. The linearized quadratic form has the correct gap $2\sqrt{\chi} - \chi$ in the range $1 < \chi < 4$, but the nonlinear intermediate-amplitude terms are not sign-definite; for $\chi > 2$, the raw quadratic deficit energy can increase. The formalized files `WholeLineChiPosL2SpectralCoercivity.lean`, `WholeLineChiPosL2DeficitEnergy.lean`, and `WholeLineChiPosL2FarLeftDecay.lean` isolate the valid shallow-basin part. The logarithmic entropy succeeds because its multiplier cancels the nonlinear mobility exactly.

Deep-hole refill. At a sufficiently deep interior minimum, locality of the elliptic resolver makes v small and the logistic reaction positive, uniformly in χ . This KPP-type refill mechanism is formalized in `WholeLineChiPosDeepHoleRefill.lean`. It controls symmetric interior holes, but it cannot stop a monotone transition layer from crossing the boundary of the moving half-line.

Front anchoring. If convergence to the selected traveling wave were already known, the far-left floor would follow. Using that convergence to prove the floor is circular in the enlarged $\chi < 4$ range; the available wave-stability assembly itself consumes far-left control or weighted closeness.

Parabolic Harnack. A Harnack inequality can propagate a positive local-mass seed into a point-wise floor. It does not create the seed. The missing estimate is precisely a uniform late-time local mass bound in a fixed moving half-line, and convective escape shows that no such seed exists without phase/speed information.

7 Lean theorem map and final conclusion

Mathematical component	Principal source file / theorem
Sharp dispersion threshold	WholeLineChiPosDispersionSharp.lean: dispersion_le_of_lt_turing, dispersion_pos_of_gt_turing
Sharp periodic entropy	WholeLineChiPosSharpEntropyDissipation.lean: sharp_entropy_production_le
Weighted resolver and entropy	WholeLineChiPosWeightedEntropyDissipation.lean: weighted_resolver_le, weighted_sharp_entropy_production_le
Canonical-orbit instantiation	WholeLineChiPosWeightedEntropyOrbit.lean: wholeLineCauchyGlobalU_weighted_sharp_dissipation
Basin-conditional convergence	WholeLineChiPosEntropyFarLeftBasinConditional.lean: uniformCoMovingLeftEquilibriumConvergence_chiPos_upto_four_basinConditional
Convective quantifier obstruction	WholeLineChiPosFloorQuantifierObstruction.lean: convectiveEscapeProfile_no_uniform_left_floor
Genuine PDE counterexample	WholeLineConvectiveEscapePDE.lean: exists_genuine_convectiveEscape_counterexample_chi_zero

The value 4 is sharp in three independent senses:

1. below 4, every linear Fourier mode decays; at 4, the mode $k^2 = 1$ is neutral; above 4, a mode grows;
2. the nonlinear entropy has the exact coercive coefficient $1 - \chi^2/16$;
3. throughout $0 \leq \chi < 4$, the localized entropy and compactness argument proves uniform far-left convergence once a persistent plateau band is supplied.

The basin/phase condition is structural: without it, a genuine front can escape convectively from a faster observation frame. The resolved theorem is therefore the basin-conditional sharp-range result, together with the convective-escape counterexample delimiting any unconditional extension.